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Masterarbeit

Master's Thesis

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Contents

1 Numerical Methods **2**
1.1 Governing Equations . . . 2
1.2 Modelling Turbulence . . . 2
1.3 Evaporation . . . 3
1.4 Multiple Flow Regimes and Interface Capturing . . . 4
1.5 Phase Interactions . . . 4
1.6 Numerical Methods . . . 5
1.7 Workflow? . . . 5

2 Subproblem: Liquid Infiltration **6**
2.1 Objective . . . 6
2.2 Validation Problems . . . 7
2.3 Conclusion? . . . 10

References **11**

1 Numerical Methods

1.1 Governing Equations

Multiphase flow is modelled in Star-CCM+ using various frameworks. The two-fluid - eulerian multiphase - framework solves a set of mass, momentum, and energy conservation equations for both the fluids to model two-phase flow. The local instantaneous conservation equations are euler averaged resulting in interfacial terms that define the influences of the phases upon each other.

$$\frac{\partial}{\partial t}\int_V \alpha_i\rho_i \,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} = \int_V \sum_{j\neq i}(m_{ij}-m_{ji})\,\mathrm{d}V + \int_V \mathrm{S}_i^\alpha\,\mathrm{d}V \tag{1.1a}$$

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \alpha_i\rho_i\mathbf{v}_i\,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\mathbf{v}_i\otimes\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} = &-\int_V \alpha_i\nabla\mathrm{p}\,\mathrm{d}V + \int_V \alpha_i\rho_i\mathbf{g}\,\mathrm{d}V\\ &+\int_{\partial V}\left[\alpha_i\left(\mathbf{T}_i+\mathbf{T}_i^T\right)\right]\cdot\mathrm{d}\mathbf{a} + \int_V \mathbf{M}_i\,\mathrm{d}V + \int_V (\mathbf{F_{int}})_i\,\mathrm{d}V\\ &+\int_V \mathbf{S}_i^\alpha\,\mathrm{d}V + \int_V \sum_{i=1}^{n}(m_{ij}\mathbf{v}_j - m_{ji}\mathbf{v}_i)\,\mathrm{d}V\end{aligned} \tag{1.1b}$$

$$\begin{aligned}\frac{\partial}{\partial t}\int_V \alpha_i\rho_i\mathrm{E}_i\,\mathrm{d}V + \int_{\partial V} \alpha_i\rho_i\mathrm{H}_i\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} = &\int_{\partial V} \alpha_i(k_{\mathrm{eff}})_i\nabla T_i\cdot\mathrm{d}\mathbf{a} + \int_{\partial V}\mathbf{T}_i\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} + \int_V \mathbf{f}_i\cdot\mathbf{v}_i\,\mathrm{d}V + \int_V \mathrm{S}_{u,i}\,\mathrm{d}V\\ &+\int_V \sum_{ij} Q_i^{(ij)}\,\mathrm{d}V + \int_V \sum_{j\neq i}(m_{ij}-m_{ji})h_i(T_{ij})\,\mathrm{d}V\end{aligned} \tag{1.1c}$$

The eulerian multiphase framework is capable of resolving dispersed and separated flow whereas the volume of fluid implementation assumes sufficient mesh resolution to resolve the interface and models separated flows. Alongside the three conservation equations, the model solves an equation for volume fraction transport.

$$\frac{\partial}{\partial t}\int_V \rho\,\mathrm{d}V + \int_{\partial V}\rho\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} = \int_V \sum_i \mathrm{S}_i^\alpha\rho_i\,\mathrm{d}V \tag{1.2a}$$

$$\begin{aligned}\frac{\partial}{\partial t}\int_{\partial V}\rho\mathbf{v}\,\mathrm{d}V + \int_{\partial V}\rho\mathbf{v}\otimes\mathbf{v}\cdot\mathrm{d}\mathbf{a} = &-\int_V \nabla\mathrm{p}\,\mathrm{d}V + \int_V \rho\mathbf{g}\,\mathrm{d}V + \int_{\partial V}\left[\left(\mathbf{T}+\mathbf{T}^T\right)\right]\cdot\mathrm{d}\mathbf{a}\\ &+\int_V \mathbf{f_b}\,\mathrm{d}V - \sum_i\int_{\partial V}\alpha_i\rho_i\mathbf{v}_i\otimes\mathbf{v}_i\cdot\mathrm{d}\mathbf{a} + \int_V \mathbf{S}_i^\alpha\,\mathrm{d}V\end{aligned} \tag{1.2b}$$

$$\frac{\partial}{\partial t}\int_V \rho\mathrm{E}\,\mathrm{d}V + \int_{\partial V}\left[\rho\mathrm{H}\mathbf{v} + \sum_i \alpha_i\rho_i\mathrm{H}_i\mathbf{v}_{d,i}\right]\cdot\mathrm{d}\mathbf{a} = -\int_{\partial V}\dot{\mathbf{q}''}\cdot\mathrm{d}\mathbf{a} + \int_{\partial V}\mathbf{T}\cdot\mathbf{v}\,\mathrm{d}\mathbf{a} + \int_V \mathbf{f_b}\cdot\mathbf{v}\,\mathrm{d}V + \int_V \mathrm{S_E}\,\mathrm{d}V \tag{1.2c}$$

$$\frac{\partial}{\partial t}\int_V \alpha_i\,\mathrm{d}V + \int_{\partial V}\alpha_i\mathbf{v}\cdot\mathrm{d}\mathbf{a} = \int_V\left(\mathrm{S}_i^\alpha - \frac{\alpha_i}{\rho_i}\frac{D\rho_i}{Dt}\right)\mathrm{d}V - \int_V \frac{1}{\rho_i}\nabla(\alpha_i\rho_i\mathbf{v}_{d,i})\,\mathrm{d}V \tag{1.2d}$$

1.2 Modelling Turbulence

Flows are generally turbulent. Turbulence is chracterised by irregular fluid motion dominated by eddies and vortices of various scales such that the energy is transferred down to the smaller scales in a cascade until it dissipates, reducing the kinetic energy of the flow and raising the internal energy of the fluid [1, 2]. Although turbulence can be modelled in various ways to various levels of resolutions, turbulence in this study has been modelled using the SST k-w model. The model is based upon the reynolds averaging of the already averaged form of the conservation equations by dividing further the flow variables into a mean and fluctuating component. The

scale of this averaging is supposed to be much larger than the averaging scales assumed in the conservation equations described in the above section [3]. This averaging process may be thought of as ensemble averaging for repeatable transient situations and leads to non-linear terms containing the fluctuations of velocities that require additional closure. The effects of turbulence through this averaging procedure introduces additional terms representing the transport of momentum and heat transfer due to turbulence. Turbulence models based on RANS equations supply a general description of turbulence production, dissipation, and transport. [4, 5, 6] In order to draw closure relations, eddy-viscosity models of turbulence assume that:

- the scalar flux vector is aligned with the mean scalar gradient in the flow
- the turbulent (eddy) viscosity relates the deviatoric reynolds stress tensor to the mean velocity gradients

These assumptions are, more often than not, invalid, owing to the fact that the physics of turbulence is different than the physics culminating into the viscous stress law.[7] [USE TURBULENT MODEELING BY MANCEAU]

The implementation of the model in Star-CCM+ can be further controlled to adjust the realisability. Realisability is the postive semi-definiteness of the reynolds stress tensor. The specification of turbulence is done using the turbulence intensity and the length scale. The inherent anisotropy in turbulence is accounted for with quadratic and cubic constitutive options in the modelling of reynolds stresses. The parameters in the model are left on their default values as they are empirically modelled.

1.3 Evaporation

The evaporation process is distinguished between occuring at saturation temperatures or not. Evaporation at saturation temperatures, known as vaporisation, is a much faster process in comparison to evaporation at normal [8]. In case of vaporisation along a heated wall, most CFD models of vaporisation treat the energy transfer by partitioning the heat flux at the wall into the latent heat of the bubble vaporisation process and the sensible heat transferred to the liquid through conduction, convection, and quencing processes. This is determined by the area fractions covered by the liquid and vapour. Correlations between the heat flux and wall superheat [what is superheat] have been made using empirical results or based on modelling of sub-processes.

The sub-processes that are primarily taken into account in such determinations are:

- number density of nucleation sites
- diameter of bubble at time of detachment from the surface
- frequency of bubble detachment

Nucleation affects the number and distribution of bubbles in the fluid. Walls, particularly so in case they're heated, are more suspectible to be nucleation sites for bubble growth as the activation energy for bubble growth is much lower. This nucleation can be modelled through the domain as homogenous or heterogenous - where nucleation sites are modelled using a presumed PDF function or be determined by empirical correlations. [9]

After a stable bubble nucleus has been formed, the growth of the bubble becomes pertinent in the evaluation of the diameter at departure and the interphase energy and mass transfer. In the case of an ideal spherical bubble undergoing expansion in a uniformly superheated liquid, the rate of growth is determined by the surface tension, liquid inertia, and the differnce in pressures inside and outside the bubble. Surface tension dominates and restricts the initial bubble growth. This initial rapid expansion of the bubble results in the formation of a thin liquid layer separating the vapour from the wall surface. This microlayer can be a significant contributor to the bubble growth and heat transfer process. As the radius increases and the liquid layer around the bubble cools due to evaporation, a decrease in vapour pressure in seen. The dynamics are controlled by thermal diffusion and inertia. Inertial effects, however, become less relevant as the bubble growth rate decreases and the bubble radius grows. The interphase energy exchange determines further bubble growth. [10]

1.3.1 Effects of low pressure and gravity conditions

The absence of significant gravitaional forces alters the dynamics of vaporisation - affecting natural convection, buoyancy driven phenomenon, and bubble dynamics. Under terrestrial gravity, convection currents develop under thermal gradients, thus bringing cooler liquid to the heated surface as the vapour bubble rises due to buyoancy. This density driven natural convection is affected in low gravity conditions, maintaining the thermal gradient. Bulk fluid movement is affected and heat and mass transfer processes shift from being convection dominated to being diffusion dominated. [11, 12]

Under low gravity conditions, vapour bubbles tend to remain near the heat source, growing signifcantly larger and coalescing with smaller bubbles. Prolonged bubble presence on the surface results in drying out of the surface and unsteady rise in surface temperatures.[13] [Mention some numbers] In conditions where the dominance of surface tension forces becomes more pronounced, the surfae tension gradients can induce fluid circulation. This is known as Marangoni convection and is observed to affect bubble growth and heat transfer. [14]

Reduced system pressure impacts the overall intensity of heat transfer. The onset of nucleate boiling also requires higher superheat values in low pressure conditions. Decrease in pressure leads to a lower number of active nucleation sites and increases microlayer thickness found at the bubble-wall interface. The local heat flux density from microlayer evaporation is impacted and the microlayer evaporation area and depletion time is increased. The system is also susceptible to the formation of a continuous vapour film at the surface due to coalescing bubbles unable to escape the surface, reducing the overall heat transfer. [15]

The affected heat transfer and bubble growth phenomenon further result in lower critical heat fluxes and heat transfer efficiency. [16, 17]

1.3.2 Vaporisation in a superheated liquid

Flash evaporation is the intense vaporisation triggered through depressurisation of superheated fluids.

1.4 Multiple Flow Regimes and Interface Capturing

Star-CCM+ in its implementation of the eulerian-multiphase framework utilises a flow regime topology based on the volume fractions of the two interacting phases. In a cell volume for increasing values of the volume fraction of the second phase, the first regime considers the second phase to be dispersed in the first phase the large scale interface modelling separated flow, and the dispersed-continuous regime respectively. To calculate the interphase interactions are composed of the weighted sum of the interaction of each flow topology regime.

The interface is resolved using the High-Resolution Interface Capturing scheme. Sharp interfaces present a discontinuity in the solution domain, causing discretisation errors. In large scale interfaces, this can lead to spurious or parasitic currents. Although often negligible, these currents can be accomodated for through the incorporation of an artificial viscosity term.

The flow at large scale interfaces is close to laminar and a boundary layer is developed. In cases of high relative velocity flows, the large gradients along this boundary layer can cause turbulent fluctuations requiring the necessity of damping the turbulence along the interface.

1.5 Phase Interactions

- drag
- surface tension
- mass transfer
- energy transfer

1.6 Numerical Methods

- gradients
- solvers - implicit unsteady, segregated flow (SIMPLE)
- Relevant non-dim numbers? (reynolds, froude, jakobs, nusselt, bond, what not)

1.7 Workflow?

2 Subproblem: Liquid Infiltration

The first subproblem deals with water infiltrating the evaporation chamber during rocket ascent. The flow system assembly comprises of the hydraulic accumulator, the evaporation chamber and the injection system subassembly that drives the liquid from the hydraulic accumulator into the evaporation chamber. The hydraulic accumulator and the evaporation chamber can be heated until lift-off and current design considerations assume the temperature of the evaporation chamber and water in the accumulator to be $368.15K$ and $353.15K$ respectively. The ambient temperature at lift-off is assummed to be $280K$ based on the remarks and considerations made in [OLD-SED]. Although insulation from the environment outside the rocket is assumed and the hydraulic accumulator will be sufficiently insulated, the evaporation chamber can cool down slightly during ascent until the start of the experiment at time, $t = 140s$, after lift-off.

The over-pressure in the hydraulic accumulator that drives the water and the absolute pressure in the evaporation chamber are assumed to be $3.5bar$ and $1bar$ respectively. Water injection stops shortly before the apogee is reached and the flow is cut-off when the absolute pressure in the evaporation chamber rises to $1.7bar$. As no flow measurements are made in the injection assembly, and the hydraulic accumulator only contains a temperature sensor, it becomes necessary to estimate the inflow conditions of the water into the evaporation chamber. Using the pressure gradient between the accumulator and the evaporation chamber, the Darcy-Weisbach equation to estimate the head loss in the pipe and accomodating for the minor head losses incurred through the following components of the injection subassembly:

- adapter to reduce the pipe diameter from D to d
- two sharp $90deg$ connectors
- and six threaded connections in the piping,

the total head loss from the accumulator to the evaporation chamber can be written as,

$$h = (1 - \frac{d^2}{D^2})\frac{v^2}{2} + 2(1.1)\frac{v^2}{2} + 6(0.08)\frac{v^2}{2} + f\frac{L}{d}\frac{v^2}{2}, \tag{2.1}$$

where L is the length of the pipe, f is the Darcy friction factor, and v is the inflow velocity of the water into the evaporation chamber. The Darcy friction factor is determined using the Haaland approximation of the Colebrook equation and the velocity is estimated by solving the energy balance for the flow system assembly,

$$\frac{\Delta P}{\rho g} = \Delta h + \frac{v}{2} + h \tag{2.2}$$

The inflow velocity, v, is found to be. With a corresponding Reynolds number, $Re =$, the initial assumption of turbulent flow is validated.

With an initial pressure of $0.03MPa$ in the evaporation chamber, it is also necessary to validate if the continuum hypothesis remains valid for the air present in the chamber. The Knudsen number relates the mean free path, λ, to the scale length of the macroscopic flow property gradients and helps determine the validity of the continuum hypothesis. The magnitude of the mean free path can be used to estimate the Knudsen number.

$$\lambda = \frac{\mu}{\rho}\sqrt{\frac{\pi m}{2k_B T}} \tag{2.3}$$

For dry air, at a pressure and temperature of $0.03MPa$ and 328 respectively, molecular mass $m = 4.815e - 26$ kg, and estimating the kinematic viscosity using Sutherland's formula, the mean free path, $\lambda \approx 2.0732e - 06m$.

2.1 Objective

The objective of the first task is to model the water injection into the evaporation chamber and extract quantities of interest in order to better estimate the initial conditions for the second task. The validity of the multiphase physics models provided by Star-CCM+ and the equations implemented therein is also evaluated against the one-dimensional Stefan problem and the two-dimensional Rayleigh-Taylor instability.

2.2 Validation Problems

2.2.1 Stefan Problem

Phase changes can be regarded as the growth of the domain of one phase accompanied by the reduction of the domain of the other phase. This growth or reduction in domain sizes can be triggered by thermal as well as non-thermal processes [shumon koga] and this representative mathematical dynamic model containing moving boundaries is known as the Stefan system. The associated problem of analysing and finding the solutions to the aforementioned model is known as the Stefan problem. [koga]. The classical Stefan problem is a one dimensional solidification or melting problem citelaterjonas, jonsson. The one-dimensional Stefan problem under semi-infinite geometry, uniform initial temperature, constant boundary temperature, and thermophysical properties constant in each phase is probably the simplest, explicitly solvable problem of phase change [lectures, uni tenessee knoxville lectures], serving as a good benchmark to validate the energy equation implemented in the VOF solver in Star-CCM+. [DOES IT ALSO SERVE AS A GOOD BENCHMARK FOR SIMPLE HEAT CONDUCTION TOO??? BECOZ TWO BIRDS ONE STONE]

Assuming that the region in space, $\Omega \subset \mathbb{R}^2$, occupied by the material is the 2D semi-infinite rectangle $\Omega = \{(x,y) \in \mathbb{R} \times \mathbb{R} : x \in [0,\infty), y \in [0,\Delta h]\}$, with the solid phase and the liquid phase occupying the regions Ω_s and Ω_l respectively such that $\Omega = \Omega_s \cup \Omega_l$. At any given time, t, the two phases are separated by the interface, $\Gamma(t)$, that marks the change in phase of the material. The boundary of the domain Ω is denoted by $\partial\Omega$. The problem space, $\mathcal{D}$, is the spatial and temporal domain occupied by the material, $\mathcal{D} = \{[t_0, t_f] \times \Omega\}$. With the heat equation governing the temperature field in the liquid region, the Stefan condition at the interface, and the respective boundary conditions, the system of equations defining the one-dimensional Stefan problem can be written as,

$$\begin{cases} \frac{\partial T}{\partial t} - \alpha \frac{\partial^2 T}{\partial x^2} & = 0, \\ \rho h_m \frac{d\Gamma(x(t),t)}{dt} + \kappa \frac{\partial \Gamma(x(t),t)}{\partial x} & = 0, \\ T(x,0) & = T_l, \\ \nabla T(x,t) \cdot \mathbf{n} & = 0, \\ T(x,0) & = T_m \end{cases} \tag{2.4}$$

- the system of equations is well-posed and has a unique solution. the evolution of the temperature field along with the evolution of the phase change interface is studied. The simulation assumes laminar and unsteady flow and solves for the solution variables implicitly using the SIMPLE scheme. The 2D region is discretised using a uniform orthogonal quadrilateral mesh of varying sizes in order to study the spatial convergence of the solution. The physical parameters for the simulation are selected according to water ice, as in [boledi et al]. A constant density $\rho = 1000\,\mathrm{kg \cdot m^{-3}}$, dynamic viscosity $\mu = 8.8871e-4\,\mathrm{Pa \cdot s}$, latent heat of fusion $100000\,\mathrm{J \cdot kg^{-1}}$, and specific heat $4200\,\mathrm{J \cdot kg^{-1}K^{-1}}$. Thermal conductivities are defined as $2.33\,\mathrm{W \cdot m^{-1}K^{-1}}$ and $0.62\,\mathrm{W \cdot m^{-1}K^{-1}}$ for the liquid and solid phases respectively. All four boundaries are assigned a no-slip shear stress specification. The boundary at $x = 0$ is assigned a constant temperature, $T = 273\,\mathrm{K}$, while the other boundaries are assumed to be adiabatic. The convection and diffusion terms are discretised using the second-order upwind scheme and the least squares method was used to interpolate the values of the gradients from the cell centres onto the face centres. The transient term is discretised using the implicit Euler-method which is first-order accurate. First-order HRIC is used to reconstruct the interface. Since Star-CCM+ sets under- and over-relaxation factors conservatively, the values are left unchanged.

The evolution of the temperature field for the simulation with cell size, $\Delta h = 5e-3$ is shown in Figure [FIG]. A clear propagation of the interface from the left to right is observed with increasing simulation times. The temperature also gradually decreases in the liquid phase from the Dirichlet boundary at $x = 0$ towards the melting temperature, T_m, at the interface after which it stays constant.

- 'This is exactly what one would expect for a velocity that is decreasing in time as is the case for the classical Stefan problem. One can also see that all simulation results converge to the analytical solution for a finer mesh and a smaller time step.'
- [ABOUT CONVERGENCE]
- [ABOUT COMPARISON OF TEMPERATURE W ANALYTICAL SOLUTION]
- [ABOUT COMPARISON OF PCI W ANALYTICAL SOLUTION]

2.2.2 Rayleigh-Taylor Instability

The interface between two superposed fluid with differing densities in unstable in cases when the heavier fluid in pushed into the lighter fluid [ref 15, p192 - Birkhoff, 1954]. This instability is known as the Rayleigh-Taylor (RT) instability. The RT instability is fairly ubiquitous, occuring at various spatial and temporal scales in nature and other fluid systems. [18] The growth of perturbations at the interface is characterised by opposing pressure and density gradients [19, 20**?**].

and the density difference is represented by the *Atwood number*, $At = (\rho_H - \rho_L)/(\rho_H + \rho_L)$. Although not encountered in its ideal form, the RT instability may occur across various natural and technological processes where it is often modified by the presence of additional processes [mention which processes?]. The principles governing the nonlinear RT instability also govern any motion of one fluid in another of different density [Tryggvason rt instability], thus serving as a good measure to study the infiltration of the evaporation chamber in the first subproblem. The instability has moreover served as a classical benchmark for the evaluation of the behaviour of two phase flows [Goulding et al.] ['WHY' imminent].

We assume that the two immiscible, viscuous, and incompressible fluids occupy the regions,

$$\Omega_i(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : (-1)^i (y - h(t,x)) > 0,\ t \geq 0\},\ i = 1,2$$

and are separated by the interface, $\Gamma(t)$ where

$$\Gamma(t) = \{(x,y) \in \mathbb{R} \times \mathbb{R} : y = h(t,x),\ t \geq 0\}$$

Using the navier stokes equations for incompressible fluids, a mixture density and viscosisty, the jump conditions at the interfaces, and the initial boundary conditions, the governing equation for the Rayleigh-Taylor instability can be written as

$$\begin{cases} \rho \frac{\partial(\mathbf{u})}{\partial t} + \rho \nabla \cdot (\mathbf{u}\mathbf{u}) & = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{g}\rho, \\ \nabla \cdot \mathbf{u} & = 0, \\ [\![S(u, p + \mathbf{g}\rho)\mathbf{n}]\!] & = \sigma\kappa\mathbf{n} + [\![\rho]\!]\mathbf{g}y \\ [\![u]\!] & = 0 \\ V & = \mathbf{u} \cdot \mathbf{n} \\ u(0) & = u_0, \\ \Gamma(0) & = \Gamma_0 \end{cases} \tag{2.5}$$

where ρ is the density, $\mathbf{u}$ is the velocity vector, p is the pressure, μ is the dynamic viscosity, and g is the gravitational acceleration.

In a system of two fluids with a perturbed interface, the heavier fluid penetrates the lighter fluid due to gravitation thereby pushing the lighter fluid upwards. The initial perturbations, generally studied using linear stability analysis, lead to an exponential growth [andrews and dalziel] and transition into non-lineartiy resulting in the rise and sink of the bubbles of lighter fluid and spikes of the heavier fluid respectively [sharp, bannerjee]. Figure [name figure] illustrates the parameter definitions and nomenclature of an evolving single-mode [need i define single mode rt?] RT instability interface.

The simulation assumer turbulent and unsteady flow, utilising the SIMPLE implicit scheme. The computational domain is discretised using an orthogonal quadrilateral mesh. The convection and diffusion terms are discretised using the second-order upwind scheme while the least squares method is again used to interpolate the values of the gradients from the cell centres onto the face centres. The VOF solver utilises the first-order HRIC scheme to reconstruct the interface. The transient term is discretised using the first-order accurate implicit Euler-method. The modified K-Omega (SST k-omega) model is used to account for turbulence and surface is treated semi-implicitly. [MENTION INITIAL CONDITIONS] The instability is simulated on a 2D rectangular domain, $[0, L_x] \times [0, 4L_x]$, where L_x is the reference length. The initial interface, located at y_0, represents a single-mode perturbation with amplitude, $A = 0.1L_x$, and wavelength, $\lambda = L_x$.

$$\Gamma_0 = 2L_x + A\cos\left(\frac{2\pi}{L_x}\right)$$

The lower limit of the high Atwood number regime requires the Atwood number to be atleast 0.5 [21]. Gravitational acceleration acts in a downward direction with a magnitude of $g = 1.0$. Periodic boundary conditions are enforced in the horizontal direction and slip wall conditions are applied at the top and bottom boundaries. The physical parameters are selected to make the simulations dynamically similar to the ones studied in [18].

According to [22], a reasonable solution to the RT instability with $At \approx 0.75$ can be obtained with a 128×512 mesh. With similar mesh resolutions being used in [18, 23] to study similar RT instabilities, in addition to a mesh convergence study being performed, the domain is discretised using a uniform orthogonal quadrilateral mesh with size $\delta_h = 0.0045$. The timestep for the simulations is selected as $\delta t = 2.5 \times 10e - 4/\sqrt{At}$ similar to [24, 18]. The following cases are studied:

Case	At	Re	σ	δ_h
1	0.5	300	$1e-5$	0.05
2	0.5	300	$1e-5$	0.007
3	0.5	300	$1e-5$	0.0045
4	0.5	300	$1e-5$	0.0025
5	0.2	1000	$1e-5$	0.0045
6	0.5	1000	$1e-5$	0.0045
5	0.5	3000	$1e-5$	0.0045
6	0.5	3000	$1e-5$	0.0045

In a system of two fluids with a perturbed interface, the heavier fluid penetrates the lighter fluid due to gravitation thereby pushing the lighter fluid upwards. The initial perturbations, generally studied using linear stability analysis, lead to an exponential growth [andrews and dalziel] and transition into non-lineartiy resulting in the rise and sink of the bubbles of lighter fluid and spikes of the heavier fluid respectively [sharp, bannerjee]. Figure **??** illustrates the parameter definitions and nomenclature of an evolving single-mode [need i define single mode rt?] RT instability interface. The evolution of RT instabilities for Reynolds numbers 300, 1000, and 3000 are depicted in figure **??**. As the solution evolves, the penetration of the spike increases and it eventually rolls up and forms two transient counter-rotating vortices due to shear forces and Kelvin-Helmholtz type instabilities [hamzelhoo], which go on to contribute to turbulent mixing and chaotic flow features [Sharp].

Figure [name figure] depicts the evolution of the interface at different times for varying density ratios as simulated in Star-CCM+. Upto high Reynolds numbers, the interface structure and its temporal evolution agree qualitatively with the results published in [hamzelhoom huang, ding]. [why does it look different at high Re]. Using the bubble and spike heights, defined as the maximum vertical distance of the spike and bubble as measured from the undisturbed initial interface, the results can be quantitatively compared to [hamzelhoo] as shown

in Figure **??**. Table **??** summarises the root mean square errors and the maximum errors between the simulated results and [hamzelhoo].

Simulations conducted for Atwood numbers very close to one also qualitatively show a similar progression and evolution of structures as the simulations in [huang]. [What happens at high atwood numbers]

[Write about mesh convergence study. WHY DO RESULTS NOT MATCH?].

- [ABOUT SPATIAL AND TEMPORAL CONVERGENCE]
- [ABOUT EVOLUTION OF INTERFACE W VARYING REYNOLDS AND VARYING ATWOOD:] Figure [FIGFIG] and figure [GIFGIF] depict the evolution of the interface at different times for varying density ratios and Reynolds numbers respectively as simulated in Star-CCM+. Upto high Reynolds numbers, the temporal evolution and the structural features of the interface agree qualitatively with the results published in [hamzelhoom huang, ding]. As the solution evolves, the penetration of the spike increases and it eventually rolls up and forms two transient counter-rotating vortices due to shear forces and Kelvin-Helmholtz type instabilities [hamzelhoo], which go on to contribute to turbulent mixing and chaotic flow features [Sharp]. [WHY DO HIGH Re NUM SIMS DEVIATE FROM EXPECTED. MENTION SALIENT FEATURES OF THE EVOLUTION] Simulations conducted for Atwood numbers very close to one also qualitatively show a similar progression and evolution of structures as the simulations in [huang]. [What happens at high atwood numbers]
- [ABOUT COMPARISON OF BUBBLE HEIGHTS W DNS DATA AND L2 AND L-INF ERRORS:] Using the bubble and spike heights, defined as the maximum vertical distance of the spike and bubble as measured from the undisturbed initial interface, the results can be quantitatively compared to [hamzelhoo] as shown in Figure **??**. Table **??** summarises the root mean square errors and the maximum errors between the simulated results and [hamzelhoo].

2.2.3 Liquid Infiltration of Evaporation Chamber

- [ABOUT CONVERGENCE]
- [ON EVOLUTION OF INTERFACE and FLOW FIELD and RESULTING HEIGHT OF WATER]
- [ON EVOLUTION OF TEMPERATURE FIELD]

2.3 Conclusion?

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